

Ilja Safronovs

R&D engineer, 3D reconstruction · Giraffe360 · Riga, Latvia

I fix what looks wrong in 3D reconstructions.

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I work on 3D Gaussian splatting from sparse panoramas at Giraffe360 (PyTorch, CUDA). Most of my work: a reconstruction looks wrong, I find out why, then I test fixes in training runs until one is good enough for production. I mentored a junior engineer, then led three engineers on a new 3D reconstruction feature; both are in production. I want to lead a team next.

EXPERIENCE

Giraffe360, R&D engineer — Riga · 2025 – present

Fixed the blurry windows, outdoor views and mirrors in Giraffe360's 3D videos (PRO camera projects). In production since June 2026. instagram.com/p/DZsNRB8o772

I found the cause, then tested fixes in controlled training runs until the haze, floaters and flicker were gone.

Our training defaults use my Gaussian pruning and relocation rules (built on Taming 3DGS).

Jan–Jul 2026, in our sparse-panorama 3DGS pipeline: influence-based pruning and relocation, building on Taming 3DGS (Mallick et al., 2024) and the team's influence tracking. It measures how much each Gaussian adds to the rendered images, then removes or moves the least useful ones. June 2026: Analytic-Splatting (Liang et al., 2024) in gsplat, with my CUDA forward and backward passes, used in a downsampling step.

The looped drone video (one flight per room) started as my hackathon idea; it is a product now. I also built the exterior orbital videos (Oct 2026). instagram.com/giraffe360/reel/Dd9ODBVIHhE

Looped drone: a special mention at the Feb 2026 hackathon (3 of my ideas were recognised out of 51); I built its camera-path algorithm. Exterior orbital videos: I built the toolkit that orients, selects and crops the outdoor scene, and the per-frame video pipeline.

Led three engineers on a new 3D reconstruction feature, now in production (Oct 2026).

I wrote most of the code, reviewed and merged the others' work, and cut the release.

Mentored a junior engineer on a new depth method; it is now in production.

Mostly his work, on denser LiDAR depth; our splat training has used it since October 2026.

Built an experiment runner driven by an AI agent, an MCP server for the team task board, and a harness for coding agents.

March 2026: an experiment runner, so every teammate can run and track training on a local GPU through an AI agent. June 2026: an MCP server (Node.js), so AI agents can read and write the team task board, with authentication; colleagues use it.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.

A Rust/wgpu viewer (May 2026) that any app can embed and control, and a multi-view comparison tool (Jan 2026).

Prototypes: mirror reflections, meshes from splats, lighter splats.

Working prototypes: correct reflections in mirrors (June 2026), mesh generation from splats, a stage that cuts the splat count for viewers (May 2026), and a fly-in video intro. Feb 2026: I made Difix and NVIDIA's fixer run on the shared GPUs, so the team could evaluate them.

Team tooling: ground-truth validation for splat quality, and one 3DGS monorepo.

July 2026: ground-truth datasets and validation tools, so every teammate can check splat quality against known ground truth. July 2026: I led the restructuring into one 3DGS monorepo, with a code-size check.

Jul–Oct 2025: a small LLM service that wrote tour narration.

It used the OpenAI API and AWS S3.

Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University, Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

Probability, statistics, stochastic analysis, Monte Carlo methods, algorithms, databases.

AI Talent Program (Ponder) Nov 2024 – Jan 2025

C++, Python, maths and AI with industry mentors; it led to the Giraffe360 role.

Riga State Classical Gymnasium, Programming and Mathematics track

State mathematics olympiads; Small Mathematics University (University of Latvia).

PROJECTS

Portfolio site: a live Gaussian splat of a real flat that I trained (gsplat).

sort_visualization: 16 sorting algorithms in C++/SFML; runs in the browser (WebAssembly).

Offline Graphing Calculator: 4th place of 27 in a school competition.

monte-carlo-finance: option pricing, volatility and risk models in Python.

load-balancing-experiments: seven experiments on DNS, queueing and load balancing.

flow: offline workout tracker in Dart/Flutter, with CI and golden-image tests.

crypto-quant-lab: 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT: neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser: finds extra RTU lectures that fit your schedule.

SKILLS

3D & vision	3D Gaussian splatting (gsplat, CUDA kernels), 3D reconstruction from sparse views, computer vision, depth densification, mesh generation, 3D camera paths, splat viewers (Rust/wgpu, three.js/Spark), PyTorch
Research	Implementing papers, controlled experiments, ground-truth evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
AI automation	LLM APIs (OpenAI), MCP servers (Node.js), experiment runners for agents, coding-agent harnesses
Tools	Linux, Docker, Git/GitLab CI, AWS S3, three.js/WebGL, Pygame, Selenium, pandas
Spoken	Latvian, English, Russian